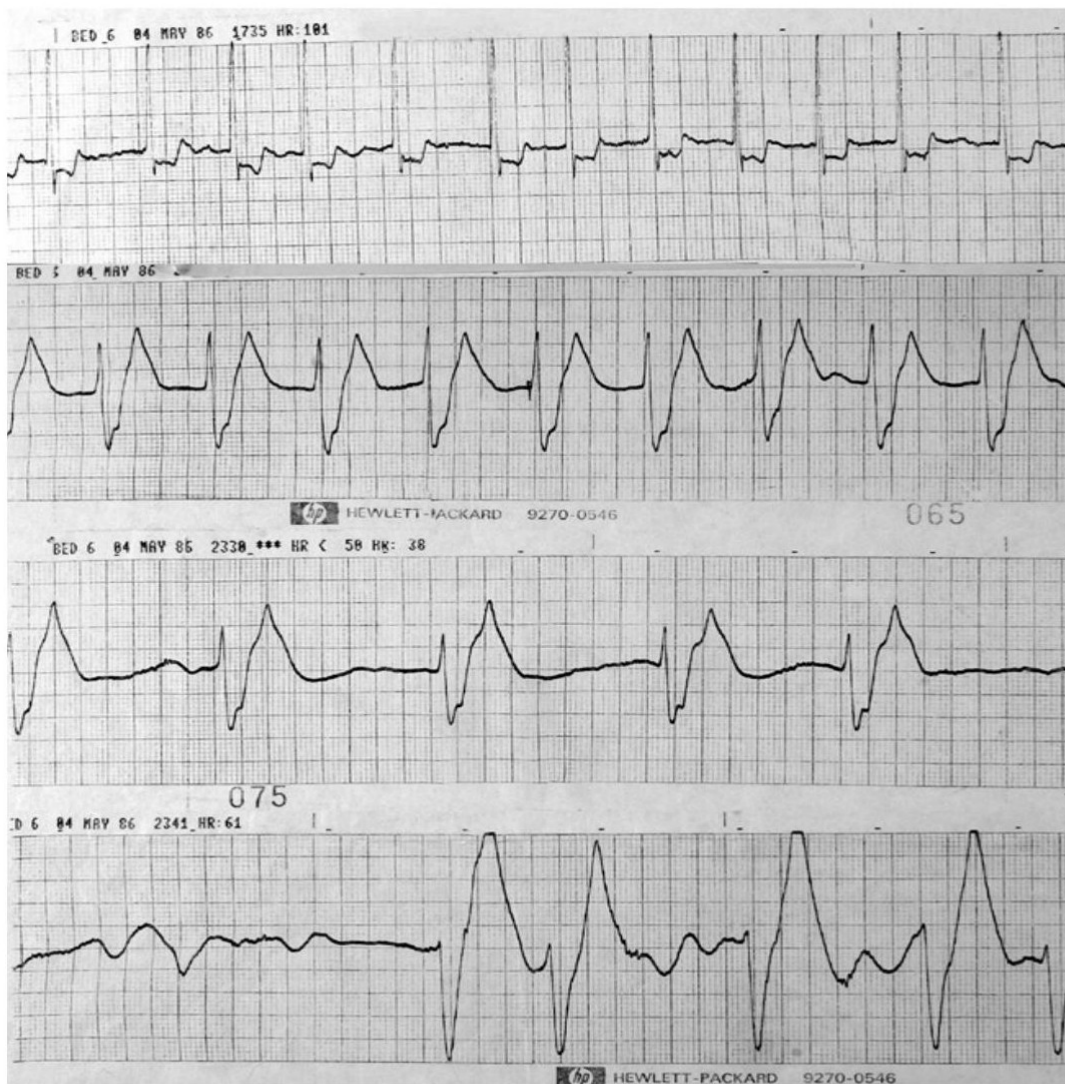


CKD & Dialysis — Companion Figures

Visual references for the CKD & Dialysis revision pack. Each figure is keyed to a ■ reference in the main .md file.

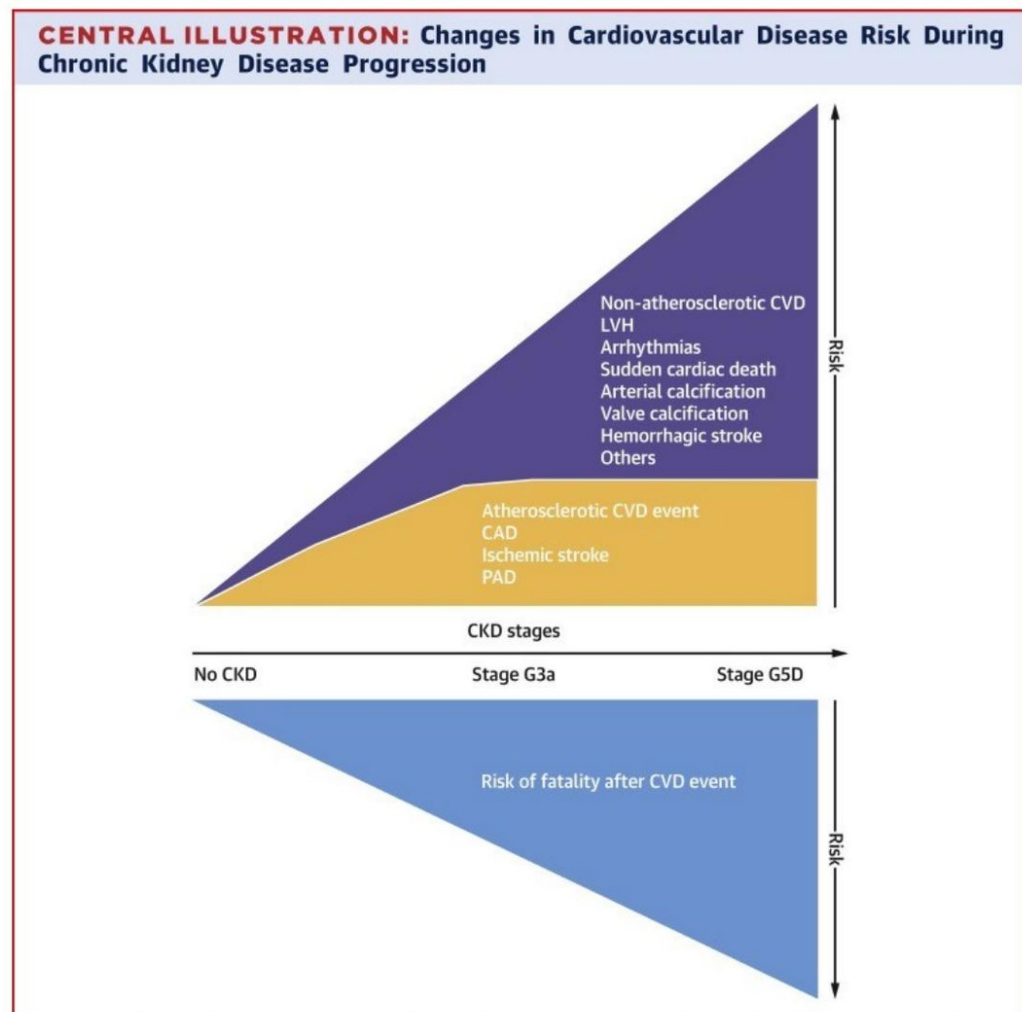


Collection of Dr Andrew Frankel

-URO-1: Urogenital homeostatic mechanism: summarise the mechanisms regulating ion/water balance and

Figure L1.2 — Progressive ECG changes in hyperkalaemia: peaked T waves → P-wave loss → wide QRS → sine wave (Collection of Dr Andrew Frankel)

Heart failure – Cardiovascular risk



Sarnak MJ, Amann K, Bangalore S, Cavalcante JL, Charytan DM, Craig JC, et al. Chronic Kidney Disease and Cardiovascular Disease: A State-of-the-Art Review. J Am Coll Cardiol. 2019;74(14):1823-38.

Figure L1.3 — Cardiovascular event risk and post-event fatality both rise with CKD stage. Non-atherosclerotic CVD (LVH, arrhythmias, calcification, sudden death) grows disproportionately. Sarnak et al. JACC 2019;74(14):1823-38.



Collection of Dr Andrew Frankel



Figure L1.1 — Case 1: Micturating cystourethrogram showing right vesicoureteric reflux — the underlying cause of long-standing CKD (Collection of Dr Andrew Frankel)

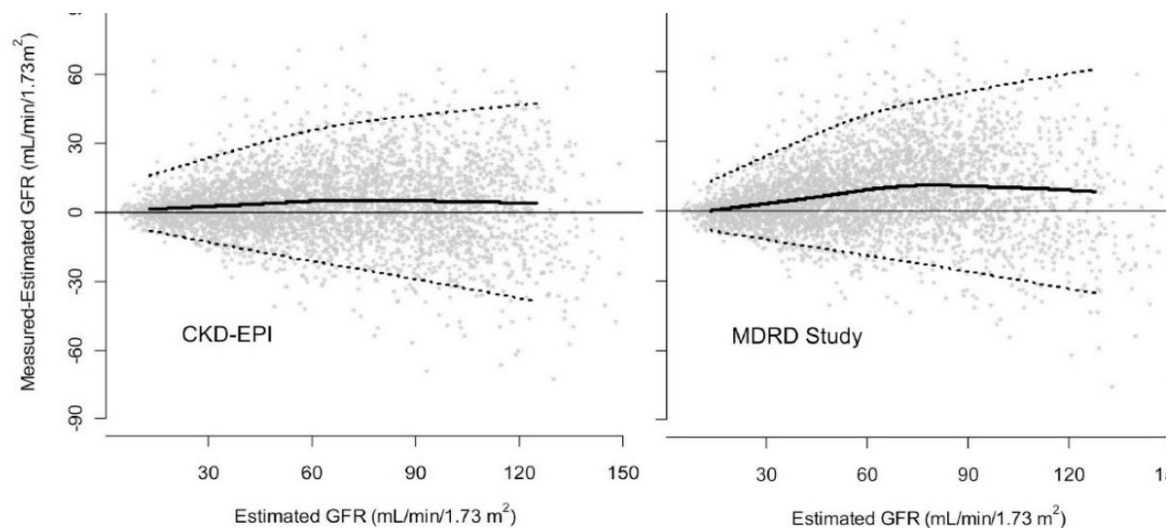


Figure L1.4 — Bland-Altman comparison: spread of measured-vs-estimated GFR widens at higher GFR. CKD-EPI (left) more accurate than MDRD (right) — NICE recommends CKD-EPI.

GFR and ACR categories and risk of adverse outcomes			ACR categories (mg/mmol), description and range		
			<3 Normal to mildly increased	3–30 Moderately increased	>30 Severely increased
			A1	A2	A3
GFR categories (mL/min/1.73 m ²), description and range	≥90 Normal and high	G1			
	60–89 Mild reduction related to normal range for a young adult	G2			
	45–59 Mild–moderate reduction	G3a			
	30–44 Moderate–severe reduction	G3b			
	15–29 Severe reduction	G4			
	<15 Kidney failure	G5			
					

Figure L1.5 — NICE CKD classification: GFR (G1–G5) × ACR (A1–A3). Risk increases along BOTH axes. G1A3 (normal GFR + heavy proteinuria) is high-risk despite normal eGFR.

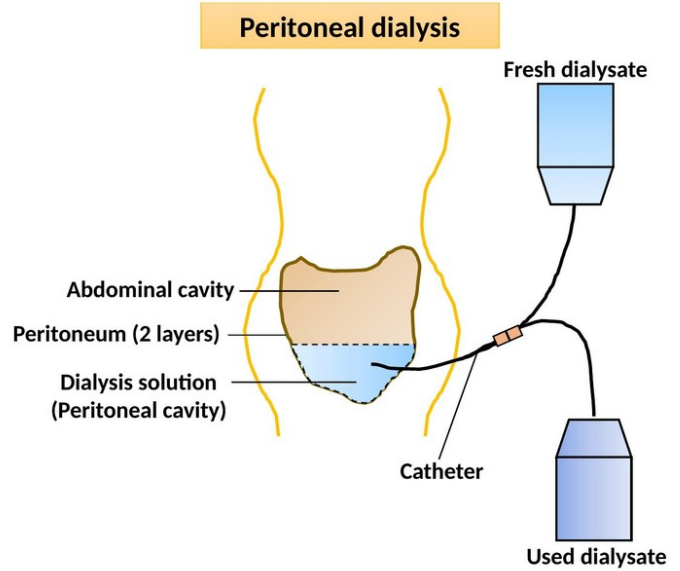
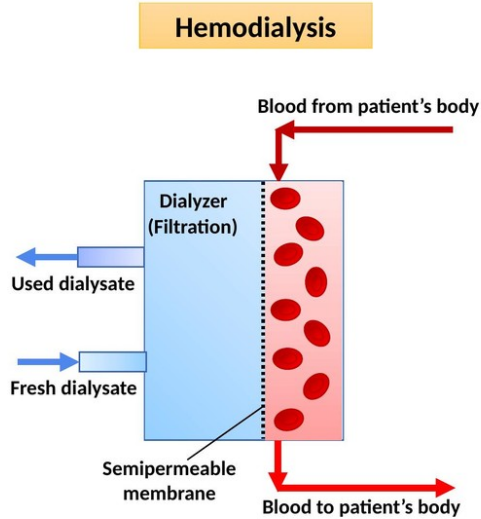


Figure L2.1 — Dialysis modalities. Haemodialysis (left): blood across a semipermeable membrane in a dialyser, counter-current to dialysate. Peritoneal dialysis (right): dialysate in peritoneal cavity, peritoneum acts as the membrane, gravity-driven exchange via catheter.

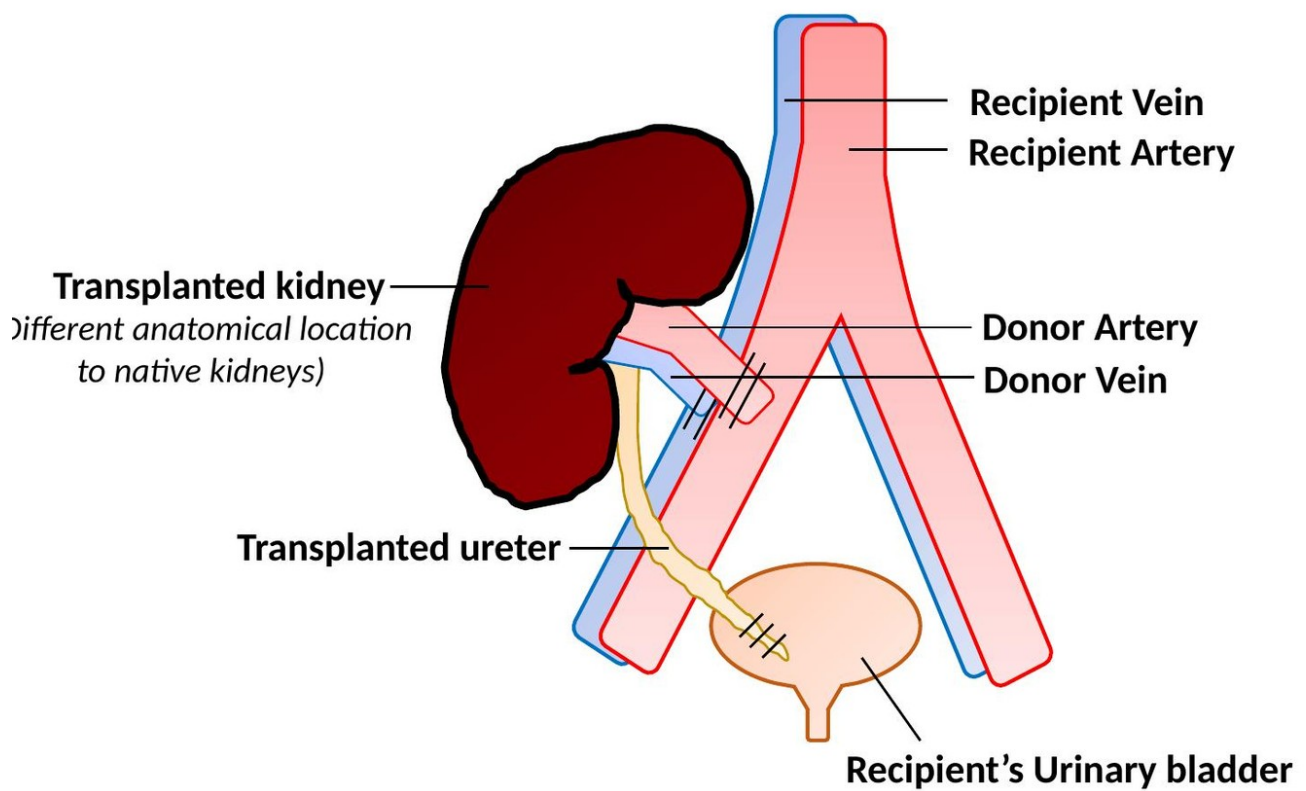


Figure L2.2 — Transplanted kidney sits in the iliac fossa (different anatomical location to native kidneys). Donor artery and vein anastomose to recipient external iliac vessels; donor ureter implants directly into recipient bladder.